Network UPS Tools Change Log

NUT project community contributors

REVISION HISTORY			
NUMBER	DATE	DESCRIPTION	NAME
2.8.5.127 2.8.5.127-127+g9b4d84125	2026-04-19	Current release snapshot of Network UPS Tools (NUT).	
2.8.5	2026-04-07	Some changes to docs and recipes, especially for parallel builds; introduced a way to build with private shared libraries to save space. Updated NUT SEMVER definition some more, added a normalization and comparison script. Some rounds of code-hardening project. Added reference RPM/DEB recipes to test NUT iterations on OBS. Numerous driver updates, some new ones introduced.	JK
2.8.4	2025-08-07	Fixed a few regressions introduced by release v2.8.3. Some changes to docs and recipes, especially for parallel builds. Updated NUT SEMVER definition some more. Some rounds of code-hardening project. Numerous driver updates, some new ones introduced.	JK
2.8.3	2025-04-07	Some changes to docs and recipes, libupsclient API and functionality. Updated NUT SEMVER definition and added scripting around it. Groundwork for vendor-defined status and INSTCMD buzzwords like "ECO". Fixed some regressions and added improvements for certain new device series.	JK
2.8.2	2024-04-01	Some changes to docs and recipes, libnutscan API and functionality. Added nutconf (library and tool). Fixed some regressions and added improvements for certain new device series.	JK
2.8.1	2023-10-31	Some changes to API, docs and recipes, in particular to simplify local builds and tests (e.g. to help end-users check if current NUT codebase trunk has already fixed an issue they see with a packaged installation). Revived NUT for Windows effort, further improved other OS integrations. NUT became reference for "UPS management protocol", Informational RFC 9271. Documentation files refactored to ease maintenance. More drivers and new driver categories introduced.	JK

REVISION HISTORY			
NUMBER	DATE	DESCRIPTION	NAME
2.8.0	2022-04-26	Change of maintainer. Many changes to API, docs (both style and content), and recipes, with a stress on non-regression test-ability, run-time debug-ability, general codebase maintainability, as well as OS integrations (notably nut-driver-enumerator for systemd and SMF service instance maintenance). Added a lot in area of CI support and documented pre-requisite package lists for numerous platforms, and CI agent set-up. Added libusb-1.x support and many new driver categories (and drivers), and daisychain device connection support. Instant commands enhanced with TRACKING to enable protocol-based waiting for completion of a particular INSTCMD or SET operation.	JK
2.7.4	2016-03-09	NUT variables namespace updated, in particular for outlet groups, alarms and thresholds, ATS devices, and battery.charger.status to supersede CHRG and DISCHRG flags published in ups.status readings. NUT network protocol extended with NUMBER type; some API changes.	AQ
2.7.3	2015-04-22	Documentation revised, including some API changes. Added NUT DDL links. NUT variables namespace updated.	AQ
2.7.2	2014-04-17	The nut-website project was offloaded into a separate repository. FreeDesktop HAL support was removed (obsoleted in GNOME consumer). Introduced nutdrv_atcl_usb driver.	AQ
2.7.1	2013-11-19	NUT source codebase migrated from SVN to Git (and from Debian infrastructure to GitHub source code hosting). jNut binding split into a separate project. Introduced libnutclient (C++ binding), al175, apcupsd-ups and nutdrv_qx drivers, Mozilla NSS support for simpler licensing than OpenSSL, and a newer apcsmart implementation. Documentation support enhanced with a spell checker, contents massively updated to reflect project changes.	CL

REVISION HISTORY

NUMBER	DATE	DESCRIPTION	NAME
2.6.5	2012-08-08	New macosx-ups driver, new implementation of mge-shut driver. NUT variables namespace updated. Docs cleaned up and revised.	AQ
2.6.4	2012-05-31	New NUT network protocol commands (LIST CLIENTS, LIST RANGE and NETVER), and socket protocol commands (ADDRANGE, DELRANGE). NUT variables namespace updated. Introduced nut-recorder tool.	AQ
2.6.3	2012-01-04	No substantial changes to documentation.	AQ
2.6.2	2011-09-15	Introduced nut-scanner tool and nut-ipmipsu driver, systemd support, and a new apcsmart implementation.	AQ
2.6.1	2011-06-01	Introduced default.* and override.* optional settings in ups.conf, an ups.efficiency report, and outlet.0 special handling.	AQ
2.6.0	2011-01-14	First release of AsciiDoc documentation for Network UPS Tools (NUT).	AQ

Contents

1 Introduction

The primary goal of the Network UPS Tools (NUT) project is to provide support for Power Devices, such as Uninterruptible Power Supplies, Power Distribution Units and Solar Controllers.

2 Very detailed Change Log

This document intends to detail the change log for relatively recent work (roughly since the source code was tracked in Git).

Note
This change log section represents git commits in range *v2.8.5..HEAD* (commits *0e051f9f6..9b4d84125*).

2.1 2026-04-18 Jim Klimov <jimklimov+nut@gmail.com>

- drivers/apc_modbus.c: addvar() the modbus_retries option [#3414]

2.2 2026-04-15 Jim Klimov <jimklimov+nut@gmail.com>

- tests/NIT/nit.sh: account SKIPPED test cases (when isTestable* says we can not) [#1711]
- tests/NIT/nit.sh: be portable, use TABCHAR instead of "\t" [#1711]
- docs/nut.dict: update for older systems
- configure.ac: work around ancient autotools without PACKAGE_URL
- Makefile.am: hijack PARMAKES_OPT to also set V=0 by default

2.3 2026-04-14 Jim Klimov <jimklimov+nut@gmail.com>

- drivers/nutdrv_qx.c, NEWS.adoc: qx_is_usb_device_supported(): only claim SUPPORTED if subdriver_command was assigned [#3410]

2.4 2026-04-14 Axel Gembe <axel@gembe.net>

- NEWS.adoc, drivers/apc_modbus.c: apc_modbus: increase default TCP response timeout to 2000 ms The default 500ms timeout value is marginal for Modbus TCP, even on a fast local network. In tests with an APC UPS, we see timeouts because of this: ``` Apr 14 19:10:44 ares apc_modbus[2002143]: _apc_modbus_read_registers: Read of 0:27 failed: Connection timed out (192.168.0.102:502) Apr 14 19:10:44 ares apc_modbus[2002143]: _apc_modbus_read_registers: Read of 0:27 failed: Invalid data (192.168.0.102:502) Apr 14 19:10:44 ares apc_modbus[2002143]: _apc_modbus_close: Closing connection Apr 14 19:10:47 ares apc_modbus[2002143]: Opened modbus successfully ``` Packet capture: ``` No. Time Source Destination Protocol Length Info 3636 2026-04-14 19:10:43.842072 192.168.0.40 192.168.0.102 Modbus/TCP 66 Query: Trans: 12622; Unit: 1, Func: 3: Read Holding Registers 3637 2026-04-14 19:10:44.018369 192.168.0.102 192.168.0.40 TCP 60 502 → 40392 [ACK] Seq=70441 Ack=11905 Win=3660 Len=0 3638 2026-04-14 19:10:44.377723 192.168.0.40 192.168.0.102 Modbus/TCP 66 Query: Trans: 12623; Unit: 1, Func: 3: Read Holding Registers 3639 2026-04-14 19:10:44.578103 192.168.0.102 192.168.0.40 TCP 60 502 → 40392 [ACK] Seq=70441 Ack=11917 Win=3648 Len=0 3640 2026-04-14 19:10:44.579449 192.168.0.102 192.168.0.40 Modbus/TCP 117 Response: Trans: 12622; Unit: 1, Func: 3: Read Holding Registers 3641 2026-04-14 19:10:44.579482 192.168.0.40 192.168.0.102 TCP 54 40392 → 502 [ACK] Seq=11917 Ack=70504 Win=63 Len=0 3642 2026-04-14 19:10:44.579604 192.168.0.40 192.168.0.102 TCP 54 40392 → 502 [FIN, ACK] Seq=11917 Ack=70504 Win=63 Len=0 ``` 3636 is the first read, 3638 is the retry after 500ms without response, both are ACKed. 3640 is the actual response to transaction 12622, which arrives 700ms after the initial packet. At this point we expect a response to transaction

12623, which is why we fail with "invalid data". To fix this we increase the default TCP response timeout to 2000 ms, which seems to be more reasonable for a device that already can take 700 ms to respond under ideal conditions. I chose a bit higher value for users who might connect the device over a higher latency network.

- NEWS.adoc, docs/nut.dict, drivers/apc_modbus.c, drivers/apc_modbus.h: apc_modbus: decode RunTimeCalibrationStatus_BF `RunTimeCalibrationStatus_BF` is explained in `MPAO-98KJ7F_R1_EN` Appendix B. This adds defines for the bitfield and decodes it into `experimental.ups.calibration.result` using a string join converter.

- NEWS.adoc, docs/nut.dict, drivers/apc_modbus.c, drivers/apc_modbus.h: apc_modbus: decode PowerSystemError_BF as alarms `PowerSystemError_BF` is explained in `MPAO-98KJ7F_R1_EN` Appendix B. This adds defines for the bitfield and decodes it as alarms in the driver.

- NEWS.adoc, docs/nut.dict, drivers/apc_modbus.c, drivers/apc_modbus.h: apc_modbus: decode BatterySystemError_BF as alarms `BatterySystemError_BF` is explained in `MPAO-98KJ7F_R1_EN` Appendix B. We were already using the bitfield for the `RB` status, this adds defines for the bitfield and decodes it as alarms in the driver.

- NEWS.adoc, docs/man/apc_modbus.txt, drivers/apc_modbus.c: apc_modbus: simplify error handling with read retries Replace the complex `_apc_modbus_handle_error` function with a simpler retry mechanism built into `_apc_modbus_read_regist` The new approach: - Retries register reads on `ETIMEDOUT` errors up to `modbus_retries` times (configurable, default 3) - On non-timeout errors or after retry exhaustion, closes the connection for reconnection on the next update cycle - Removes the platform-specific (WIN32/POSIX) timeout detection and the flush-based recovery that didn't work anyway (flush is already done in `_apc_modbus_reopen` upon reconnection) Also adds a `modbus_retries` driver option to configure the number of read retry attempts, and improves logging for connection open/close events. This change was inspired by a patch by @marcan to do the same and from testing the behaviour of `apcupsd`, noticing that on my USB unit it times out on read and only succeeds on the first retry.

- NEWS.adoc, drivers/apc_modbus.c: apc_modbus: always zero terminate string join `_apc_modbus_string_join` did not zero terminate the result in every case, like when all values are `NULL`.

- drivers/apc_modbus.c: apc_modbus: bump driver version to 0.20

2.5 2026-04-13 Jim Klimov <jimklimov+nut@gmail.com>

- conf/upsd.conf.sample, docs/man/upsd.conf.txt: heads-up about MAXCONN and ulimit [#3365]

- NEWS.adoc: remove duplicated paragraph (silent merge conflict)

- clients/upsmon.c, clients/upsrw.c, common/common.c, drivers/adelsystem_cbi.c, drivers/generic_modbus.c, drivers/main.c, drivers/powerman-pdu.c, drivers/snmp-ups.c, drivers/tripplite_usb.c, tools/nut-scanner/nutscan-serial.c: **.c: fix use of non-const char** with strstr() and strchr() return values [#823]

- docs/man/Makefile.am: fix possible non-delivery of NUTSCAN_UPSLOG_SET_DEBUG_LEVEL_DEPS pages [#3378, #3379]

- docs/man/upscli_init.txt: update wording and markup for pre-existing features [#3331]

- NEWS.adoc: announce the adjusted markup format

2.6 2026-04-12 Jim Klimov <jimklimov@gmail.com>

- scripts/perl/UPS/Nut.pm: warn if CERTVERIFY and custom CA options are enabled on Darwin platform [#3404]

- scripts/perl/UPS/Nut.pm: align closer with can_ssl [#1711]

- scripts/perl/UPS/Nut.pm, scripts/perl/test_nutclient.pl, tests/NIT/nit.sh: scripts/perl/*, tests/NIT/nit.sh: streamline passing NUT_DEB (or defaulting to no debug) [#1711]

- tests/NIT/nit.sh: neuter CERTVERIFY on darwin for now [#3404]

- tests/NIT/nit.sh: centralize definition of PERL_OPTS [#1711]

- scripts/perl/UPS/Nut.pm: allow SSL cert validation for IP addresses as host names [#1711]

- scripts/perl/UPS/Nut.pm: pass only defined args to IO::Socket::SSL→start_SSL() [#1711]
- scripts/perl/UPS/Nut.pm: support debugging messages of IO::Socket::SSL [#1711]
- tests/NIT/nit.sh: report the non-default NIT_CASE being handled [#1711]

2.7 2026-04-12 Jim Klimov <jimklimov+nut@gmail.com>

- scripts/perl/UPS/Nut.pm, scripts/perl/test_nutclient.pl: scripts/perl/UPS/Nut.pm: now we can pass SSL debug level as a number [#1711]
- scripts/perl/UPS/Nut.pm: fix to loading IO::Socket::SSL via eval with quotes not braces [#1711] It is then evaluated at run-time as code reaches that line (or not - conditionals), not at interpretation time where it fails too early. Also use "1;" in the end as a tidy practice.
- clients/nutclient.cpp: work around lack of threads in some mingw versions [#3402]

2.8 2026-04-11 Jim Klimov <jimklimov+nut@gmail.com>

- m4/ax_c_pragmas.m4, clients/nutclient.cpp: introduce HAVE_PRAGMA_GCC_DIAGNOSTIC_IGNORED_DEPRECATED_DECL to implement methods for deprecated "master" operation [#840, #3402]

2.9 2026-04-10 Jim Klimov <jimklimov+nut@gmail.com>

- NEWS.adoc: announce updates to client binding libraries [#3402]
- scripts/perl/test_nutclient.pl: avoid "Can't call method "Error" on an undefined value" messages in failed tests; clearly say that $nut is no more [#1711]
- scripts/perl/UPS/Nut.pm: when IO::Socket::SSL is not available, report the error with "FEATURE-NOT-SUPPORTED" in text [#1711]
- tests/NIT/nit.sh: setenv_ssl_perl(): check if IO::Socket::SSL is available, neuter NUT_FORCESSL based on that [#1711]
- scripts/perl/test_nutclient.pl: support USESSL=undef if no value provided in env [#1711]
- scripts/perl/UPS/Nut.pm: support USESSL=undef, check that we can_ssl and follow up with that [#1711]
- scripts/perl/UPS/Nut.pm: _initialize(): include previously reported error in SSL setup fail messages [#1711]
- scripts/perl/UPS/Nut.pm: apply indentation consistently (3 styles were intermixed)
- scripts/perl/UPS/Nut.pm, scripts/perl/test_nutclient.pl: support NUT_SSL envvar toggle to disable attempts at SSL altogether [#1711]
- scripts/perl/UPS/Nut.pm, scripts/perl/test_nutclient.pl: SSL_ca_path should be defined in args by caller, not by a hack in production code [#1711]
- scripts/perl/UPS/Nut.pm: STARTTLS: debug-print %arg list that is passed to the SSL library [#1711]
- scripts/perl/UPS/Nut.pm: use CAPATH to provide SSL_ca_path, not SSL_ca_file [#1348, #1711]
- scripts/python/module/PyNUT.py.in: retry SSL handshake if it timed out, a few times [#3401]
- server/netssl.c: net_starttls(): report which SSL backend is in place (if any) [#3331]
- scripts/python/module/PyNUT.py.in: isValidProtocolVersion(): stringify and strip the "result" before matching in regex [#1349]
- clients/Makefile.am: bump libupsclient ABI version
- tests/NIT/nit.sh: add a note about jNut using this script [#1711, #1350]

- clients/nutclient.cpp, clients/nutclient.h, clients/nutclientmem.cpp, clients/nutclientmem.h, tests/cpputest-client.cpp: clients/nutclient{,mem}.{h,cpp}, tests/cpputest-client.cpp: remember if we enabled TRACKING so it is really done once, confirm with isTrackingModeEnabled() [#656]
- scripts/perl/test_nutclient.pl: fix DeviceLogin⇒ListClient for "admin" without a role [#1711]
- scripts/perl/UPS/Nut.pm: _send(): reset the error buffer before networking away [#1711, #1348]
- scripts/perl/UPS/Nut.pm: we can only authenticate once per connection, skip other attempts [#1711, #1348]
- docs/download.txt: publish jNut-1.1 [#1350]
- NEWS.adoc: announce updates to client binding libraries [#3402]
- appveyor.yml: post only one (custom) notification per build [#1400]
- docs/man/Makefile.am: hush "cat" errors in LIBNUTCLIENT_TCP_DEPS_SUB recipe

2.10 2026-04-09 Jim Klimov <jimklimov+nut@gmail.com>

- scripts/perl/test_nutclient.pl: fix reporting of @failed methods [#1711]
- scripts/perl/UPS/Nut.pm: GetTrackingResult(): fix parsing results of GET TRACKING [#1348]
- tests/NIT/nit.sh: when we ask for specific NIT_CASE values (cppnit, python, perl, nutscanner) the lack of prerequisites should be a FAILURE [#1711]
- tests/NIT/nit.sh: introduce tests for PERL [#1348, #1711]
- message, so we can parse it [#1348, #1711]
- scripts/perl/test_nutclient.pl: fix access to possibly undefined envvars [#1348, #1711]
- scripts/perl/UPS/Nut.pm: fix error messages when Authenticate() and Login() fail [#1348, #1711]
- scripts/perl/UPS/Nut.pm, scripts/perl/test_nutclient.pl: scripts/perl/test_nutclient.pl: fix test case setup, update comments [#1711, #1348]
- scripts/perl/UPS/Nut.pm: fix string work with TRACKING logic
- scripts/perl/UPS/Nut.pm: change commands that address an UPS to optionally accept the UPS name as argument, only using
- scripts/python/module/test_nutclient.py.in: fix typo in debug traces
- assignments into debug() calls [#1348, #1711]
- COPYING, scripts/HP-UX/nut.psf.in, scripts/Makefile.am, scripts/obs/debian.libups-nut-perl.install, scripts/perl/{ ⇒ UPS}/Nut.pm, scripts/perl/test_nutclient.pl: scripts/perl/test_nutclient.pl: introduce tests, move Nut.pm to UPS subdir [#1348, #1711]
- scripts/perl/Nut.pm: call StartTLS() from _initialize(), and use isValidProtocolVersion() method to actively ping a connection after STARTTLS claimed success (but handshake might be broken in fact) [#3387, #1348]
- clients/nutclient.cpp, clients/upsclient.c, scripts/python/module/PyNUT.py.in: Client libraries (C, C+\+, Python): use isValidProtocolVersion() method to actively ping a connection after STARTTLS claimed success (but handshake might be broken in fact) [#3387]
- clients/nutclient.cpp, clients/nutclient.h, clients/nutclientmem.cpp, clients/nutclientmem.h, clients/upsclient.c, clients/upsclient.h, scripts/perl/Nut.pm, scripts/python/module/PyNUT.py.in: Client libraries (C, C+\+, Python, PERL): introduce an isValidProtocolVersion() method to actively ping a connection [#3387]
- clients/nutclient.cpp, clients/nutclient.h, clients/nutclientmem.cpp, clients/nutclientmem.h: clients/nutclient{,mem}.{h,cpp}: becomeSecondary() was overkill, there is no such verb in NUT protocol (LOGIN already reaches that role level)
- scripts/python/module/PyNUT.py.in: refactor FSD() with an explicit becomePrimary() method

- scripts/perl/Nut.pm: becomeSecondary() was overkill, there is no such verb in NUT protocol (LOGIN already reaches that role level)
- scripts/perl/Nut.pm: add warnings toggle
- scripts/perl/Nut.pm: fix typos in comments
- scripts/perl/Nut.pm: rename primary()⇒becomePrimary, add becomeSecondary() for completeness [#1348]
- clients/nutclient.cpp, clients/nutclient.h, clients/nutclientmem.cpp, clients/nutclientmem.h: clients/nutclient{,mem}.{h,cpp}: rename primary()⇒becomePrimary, add becomeSecondary() for completeness [#656]
- tests/cpputest-client.cpp: add test with SET VAR TRACKING [#1711, #656]
- scripts/python/module/test_nutclient.py.in: add test with SET VAR TRACKING [#1711, #1349]
- scripts/perl/Nut.pm: fix typos in comments
- docs/download.txt: refer to new release of jNut-1.0 (and the following snapshot evolution)
- tests/NIT/nit.sh: do not unset DEBUG_SLEEP when we generate configs [#1711] We might actually want more of those, with test case passwords revealed, etc.

2.11 2026-04-08 Jim Klimov <jimklimov+nut@gmail.com>

- scripts/python/module/PyNUT.py.in: TrackingID class: include timestamp tracking and age reporting [#1349]
- scripts/perl/Nut.pm: TrackingID class: include timestamp tracking and age reporting [#1348]
- clients/nutclient.cpp, clients/nutclient.h, clients/nutclientmem.cpp, clients/nutclientmem.h: clients/nutclient{,mem}.{h,cpp}, tests/*.cpp: implement waiting for TRACKING result if requested [#656]
- clients/nutclient.h, clients/Makefile.am: Implement TrackingID as a separate class, including timestamp tracking and age reporting [#656]
- scripts/python/module/PyNUT.py.in: Implement TrackingID as a separate class [#1349]
- scripts/python/module/PyNUT.py.in: fix syntax [#1349]
- scripts/perl/Nut.pm: Implement TrackingID as a separate class [#1348]
- scripts/perl/Nut.pm: fix syntax [#1348]
- scripts/python/module/PyNUT.py.in: revise AI creations, complete the TRACKING\+waiting dialog support [#1349]
- scripts/perl/Nut.pm: revise AI creations, complete the TRACKING\+waiting dialog support [#1348]
- scripts/perl/Nut.pm, scripts/python/module/PyNUT.py.in: AI PoC: implement the remainder of NUT Networked protocol (as of NUT v2.8.5) for Python and Perl bindings
- scripts/perl/Nut.pm: clean up trailing white-space
- tests/NIT/nit.sh: simplify with NIT_CASE=generatecfg_sandbox support [#1711]
- tests/NIT/nit.sh: allow injecting WITH_SSL_CLIENT, WITH_SSL_SERVER, WITH_SSL_SERVER_CLIVAL (e.g. to test against older packaged builds) [#1711]
- .github/workflows/05-codeql.yml: drop duplicate init stage
- configure.ac: hush clang-21 "-Wc11-extensions" warnings about system headers
- docs/download.txt: fix NOTE markup for N4W 7Zip size

2.12 2026-04-07 Jim Klimov <jimklimov+nut@gmail.com>

- docs/maintainer-guide.txt: revise instructions for post-release steps (website, etc.)
- docs/download.txt: newly refer to nut-X.Y.Z-docs.tar.gz archives
- docs/docinfo.xml.in: fix release date of v2.8.5 and move its tag to be above v2.8.4
- NEWS.adoc, UPGRADING.adoc, docs/docinfo.xml.in: Revert "NEWS.adoc, UPGRADING.adoc, docs/docinfo.xml.in: finalize text before NUT v2.8.5 release" This reverts commit 784219a3dc935e53a1273063c591bc6dd949590b: finally released NUT v2.8.5, preparing for v2.8.6 cycle now.

2.13 2026-04-06 Jim Klimov <jimklimov+nut@gmail.com>

- common/wincompat.c: syslog(), send_to_named_pipe(): debug-log when we failed to open existing NAMED_PIPE and so move on [#3302]
- common/wincompat.c: log errors after methods faults via upslog_with_errno() not plain upslogx() [#3302]
- common/wincompat.c: log progress through pipe*() methods [#3302, #3368]
- common/wincompat.c: fix send_to_named_pipe() to measure strlen(data) once [#3302]
- common/wincompat.c: fix getpass() to measure strlen(wincompat_password) once [#3302]
- common/wincompat.c: fix filter_path() to measure strlen(source) once [#3302]
- common/wincompat.c: tcflush(): fix bitwise OR to boolean OR